

Filip Nový

AI / ML Engineer — Retrieval Systems & Applied Machine Learning

Kolding, Denmark | EU Citizen (Czech) | Open to relocation
filip.novy@hotmail.com | +45 28 56 48 24 | DOB: 24 March 2001
LinkedIn | GitHub | filipnovy.dk



Profile

AI engineer who has built and deployed a production RAG system end-to-end: from PDF ingestion and knowledge graph construction through retrieval pipeline design to cloud deployment and rigorous evaluation. Scientific paper (under review) demonstrating measurable improvements in answer faithfulness through a controlled experimental methodology. Prior industry experience at T-Mobile Czech Republic building dashboards and data pipelines adopted by regional sales teams. Looking for ML, NLP, or AI engineering roles where end-to-end ownership and measurable evaluation matter.

Technical Skills

Languages	Python (scikit-learn, PyTorch, LangChain, FastAPI, Flask, Pydantic), SQL (Oracle, MS SQL, PostgreSQL), R, Java (basics)
ML & NLP	Transformers, dense embeddings & cross-encoder reranking, RAG / GraphRAG, vector databases (ChromaDB), knowledge graphs (Neo4j), MultiQuery retrieval, structured LLM output (Pydantic, instructor)
LLM Ops	Cloud & local LLM inference, API integration (OpenAI and others), prompt engineering, RAGAS evaluation (faithfulness, context precision, context recall), custom evaluation frameworks, agentic workflows
MLOps & Tools	Git, Docker (basics), Streamlit, Power BI, Plotly / Dash, DigitalOcean
Methods	Deep learning, statistical testing, linear programming, experimental design, model evaluation & explainability

Publications

Not All Context Helps: Isolating Graph Expansion and Community Summaries in Scientific RAG \ Under review at KONVENS 2026.

Controlled evaluation of a retrieval-augmented generation system serving 879 scientific papers. Isolated the contribution of each pipeline component through a factorial ablation with statistical validation, identifying which architectural decisions improve answer quality and which introduce noise. The optimized configuration achieved **+4.8%** faithfulness over baseline. Methodology applicable to any production RAG system requiring evidence-based design choices.

Education

MSc Data Science

Sept 2024 — June 2026 (expected)

University of Southern Denmark (SDU), Kolding

Selected coursework (grade 12 / A): Deep Learning, Applied Machine Learning, Natural Language Processing, Data Visualization

Volunteer student tutor (2025): mentored incoming students across 30+ nationalities; recognized in a written recommendation by Programme Coordinator Søren Brolund Ulriksen.

Thesis — Domain-Specific Chatbot for Algae Research: A Hybrid GraphRAG Approach

Supervisor: Tariq Yousef | Defense: June 2026

- Built and deployed a production RAG system over ~880 peer-reviewed PDFs, handling the full pipeline: PDF extraction with OCR fallback, recursive chunking (~32k chunks), multi-query reformulation, dense vector retrieval with cross-encoder reranking, and cloud deployment on DigitalOcean.
- Constructed a knowledge graph (~60k entities, 100k+ edges) using structured LLM extraction with Pydantic schema validation, processing 31,738 chunks at approximately **\$8 total API cost** through cost-optimized model routing.
- Designed a two-stage summarization pipeline that reduced token costs by an order of magnitude relative to standard approaches, routing document compression through a small local model before cloud-based community-level synthesis.

- Deployed as a live Streamlit application (algaebot.filipnovy.dk) with knowledge graph visualization, retrieval transparency, and automatic graceful fallback when graph infrastructure is unavailable.

BSc Applied Computer Science

2020 — 2024

Prague University of Economics and Business (VŠE)

Thesis: *An Analysis of the Current State of Research on Artificial General Intelligence* (grade A).

Erasmus exchange semester in Poland.

Professional Experience

Data Analyst Trainee

April 2023 — August 2024

T-Mobile Czech Republic, Prague

- Designed and shipped Power BI dashboards for financial forecasting and sales-lead prioritization that were adopted by regional sales teams across the Czech Republic, replacing manual Excel workflows.
 - Engineered SQL data pipelines against Oracle and MS SQL sources to consolidate fragmented customer and revenue data into a single analytics layer, reducing recurring reporting effort.
 - Served as the translation layer between non-technical business stakeholders and the analytics team, scoping requirements and communicating model assumptions and limitations.
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Selected Projects

Deep Learning Portfolio | *PyTorch, TensorFlow*

A curated GitHub repository of from-scratch implementations covering CNNs, RNNs/LSTMs, autoencoders, variational autoencoders, GANs, and Transformers, alongside the underlying optimization machinery (SGD variants, backpropagation). Built across deep-learning coursework and self-study; serves as a portable reference for the architecture families I work with.

Vision Transformer for Plant Disease Classification | *PyTorch*

Fine-tuned a Vision Transformer with custom attention heads on 87,000 leaf images spanning 38 disease classes, achieving **99.77%** validation accuracy. Oral exam grade: 12 (highest).

Quantitative Trading System in a Virtual Economy | *Python, Flask, Survival Analysis, Linear Programming*

Built an end-to-end trading advisor ingesting a public game-economy API; a survival model estimated order-fill probabilities from transaction history, feeding a linear program that allocated capital across items to maximize expected return. Ranked in the **top 0.1%** (380th of 460,000+ accounts) by actively traded capital.

Primate Diet Classification from Dental Morphology | *R, LDA, QDA*

Multivariate-statistics project classifying probable diets of fossil primates from tooth-measurement data using Linear and Quadratic Discriminant Analysis, with model comparison and cross-validated accuracy reporting.

Music Trends Across the Decades | *Plotly, Dash*

Interactive web dashboard aggregating Spotify audio-feature data to tell a narrative about how popular music has evolved over time; capstone for the Data Visualization course.

Languages

Czech (Native) | English **C2** (Cambridge CPE) | German **B1-B2** | Danish **A2**

References

Tariq Yousef — Thesis Supervisor, University of Southern Denmark. **Søren Brolund Ulriksen** — Programme Coordinator, MSc Data Science, University of Southern Denmark. Letters of recommendation available on request.